Leonardo Mattos Martins

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Education

Boston University, Cum Laude BS in Computer Engineering (3.6/4.0 GPA) May 2026

- **Achievements:** Concentration in Machine Learning, Minor in Data Science, Dean's List
- **Relevant Coursework:** Algorithms & Data Structures, High-Performance GPU Programming, Deep Learning, Sequential Decision Making, AI For Science, Full-Stack Programming, Optimization Theory, AI Ethics, Cybersecurity

Skills and Technologies

Languages (in order of proficiency): Python, C/C++, Java, Javascript, MATLAB, Rust, Go, SQL, Perl
AI & Machine Learning: PyTorch, Pandas, TensorFlow, JAX, Scikit-learn, TensorRT, Llama, HuggingFace, Tinker, CUDA
Tools & Infrastructure: Git, Copilot, Docker, CI/CD, Linux, REST, Kubernetes, MongoDB, PostgreSQL, Agile/Scrum

Projects

RL & SFT for Game-Theoretic Reasoning *[localminimaclub]* Apr 2026

- Led a team of 3 graduate researchers (selected from 50+ candidates) to architect a novel evaluation of backward induction, achieving a 27.2% improvement in logical depth for open-weight models
- Constructed a hybrid training pipeline combining SFT and GPO/DPO on 3B–7B parameter models (Llama/Mistral) via the Tinker HPC cluster, optimizing for strategic reasoning in multi-turn games
- Created a synthetic data generation engine using chain-of-thought (CoT) prompting to create 10k+ game-tree trajectories, reducing logical hallucinations by 40% in zero-shot game-theoretic benchmarks

AI4Science Jet Generation Challenge *[AI4Science-JetGenerationChallenge]* Mar 2026

- Formulated a conditional flow-matching model using RK2 integration and masked set-aware attention, achieving a 0.0382 W1 score and 1st place in an AI4Science competition amongst 150 applicants, being commended on technical prowess
- Built a robust training pipeline featuring FiLM conditioning, EMA stabilization, and AdamW cosine scheduling to enable stable convergence on variable-length particle clouds
- Refined permutation-invariant architectures for 3D physics simulation, achieving a 0.00092 W1 score on gluon jets through zero-masked velocity prediction and token padding

Vulnerability Detection *[Stack-Underflow]* Apr 2025

- Led a team to build a large-scale dataset of 100K+ Stack Overflow code snippets and benchmarked three vulnerability detection methods: Claude 4 Opus, fine-tuned CodeBERT, and a regex + static analysis baseline
- Elevated vulnerability classification accuracy by 18% over zero-shot LLMs by fine-tuning CodeBERT on labeled security datasets and optimizing for precision and recall
- Expanded detection coverage by 30% by developing a custom static analysis engine using AST parsing to identify insecure API calls, input validation issues, and injection-prone code patterns

Experience

AI Engineering & BizOps Intern, SumUp – São Paulo, BR Jun 2025 – Aug 2025

- Saved 1.5 FTEs in operational effort by streamlining 18 manual workflows through Python-based agents (RAG/MCP) and FastAPI integrations decreasing ticketing workload by 15 hours per week
- Accelerated AI adoption by 40% in the São Paulo office by delivering technical lectures on LLM APIs and establishing company-wide protocols for responsible AI integration
- Collaborated with VP of Engineering to prototype agentic AI capabilities using LangChain, resulting in a functional PoC adopted by R&D team for document parsing and reducing manual entry by 15 hours/week

AI & Automation Intern, Doctor Joe Technology – São Paulo, BR May 2024 – Aug 2024

- Processed over 5,000 hours of audio data daily by automating 100% of data throughput through a robust Python pipeline integrated with Deepgram and OpenAI APIs
- Reinforced data-driven decision making for 2 enterprise clients by leveraging GPT-4-mini to generate executive summaries, increasing client reporting speed by 40%
- Decreased manual data entry workload for sales team by 90% by translating complex business logic into automated RPA workflows using ElectroNeek